5.2 The impacts of climate change

The impacts of climate change affect all inhabitants of Earth: humans, plants, and animals. One of the most noticeable impacts of climate change on a global scale is the effects of increased temperatures. Since 1880, the global average temperature has risen by 1.2 °C. 2011–2020 was the warmest decade on record. The climate stripes in Figure 5.8 show how global average temperatures have increased over the last two centuries (1850–2017). Each stripe represents a year. Blue stripes show cooler than average years whereas red stripes show warmer than average years. This recent increase in global average temperatures has had a number of impacts on Earth's climate.

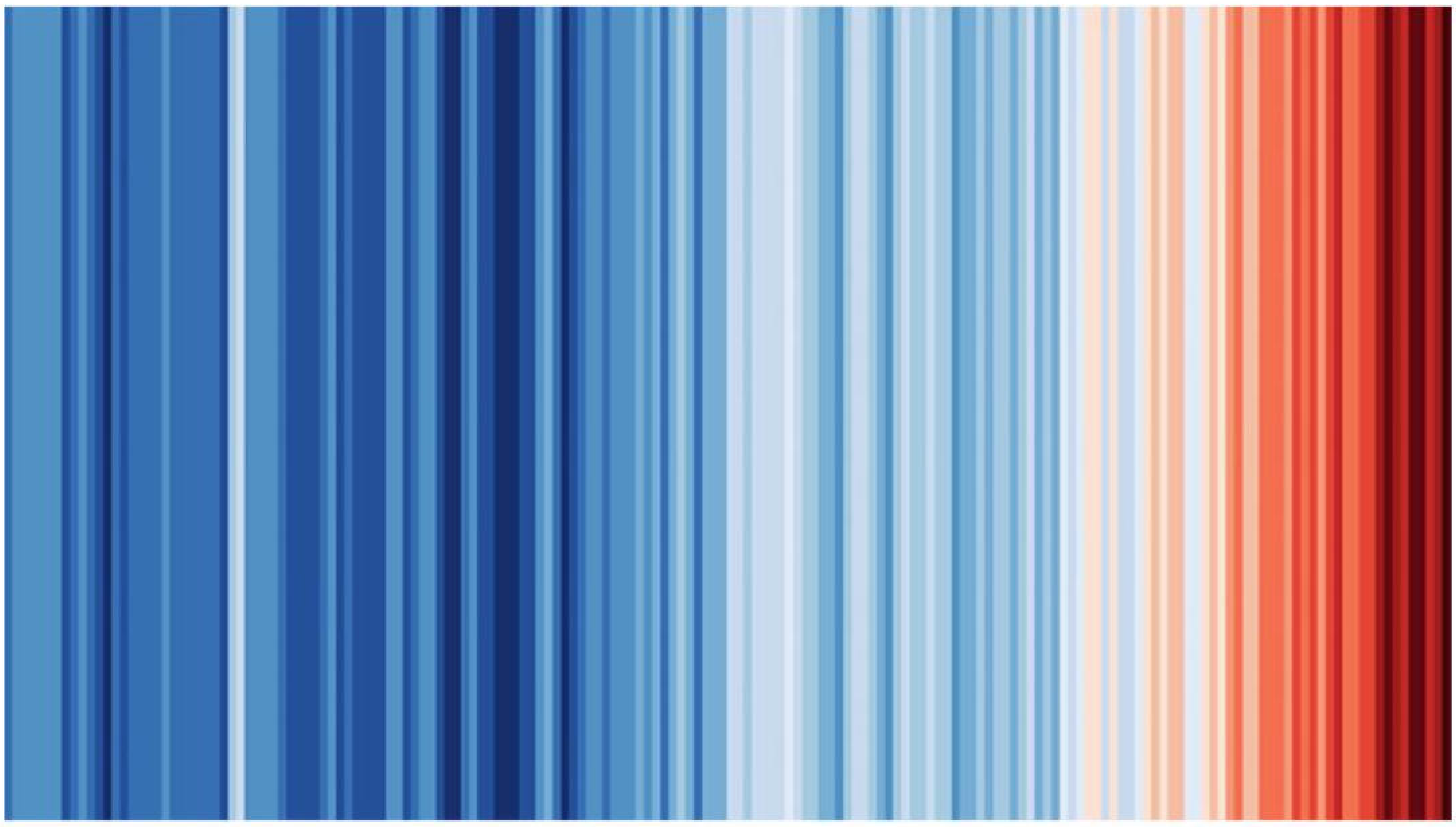


Figure 5.8: Climate stripes visual, representing the global temperature rise in the last two centuries

Sea level rise and thermal expansion

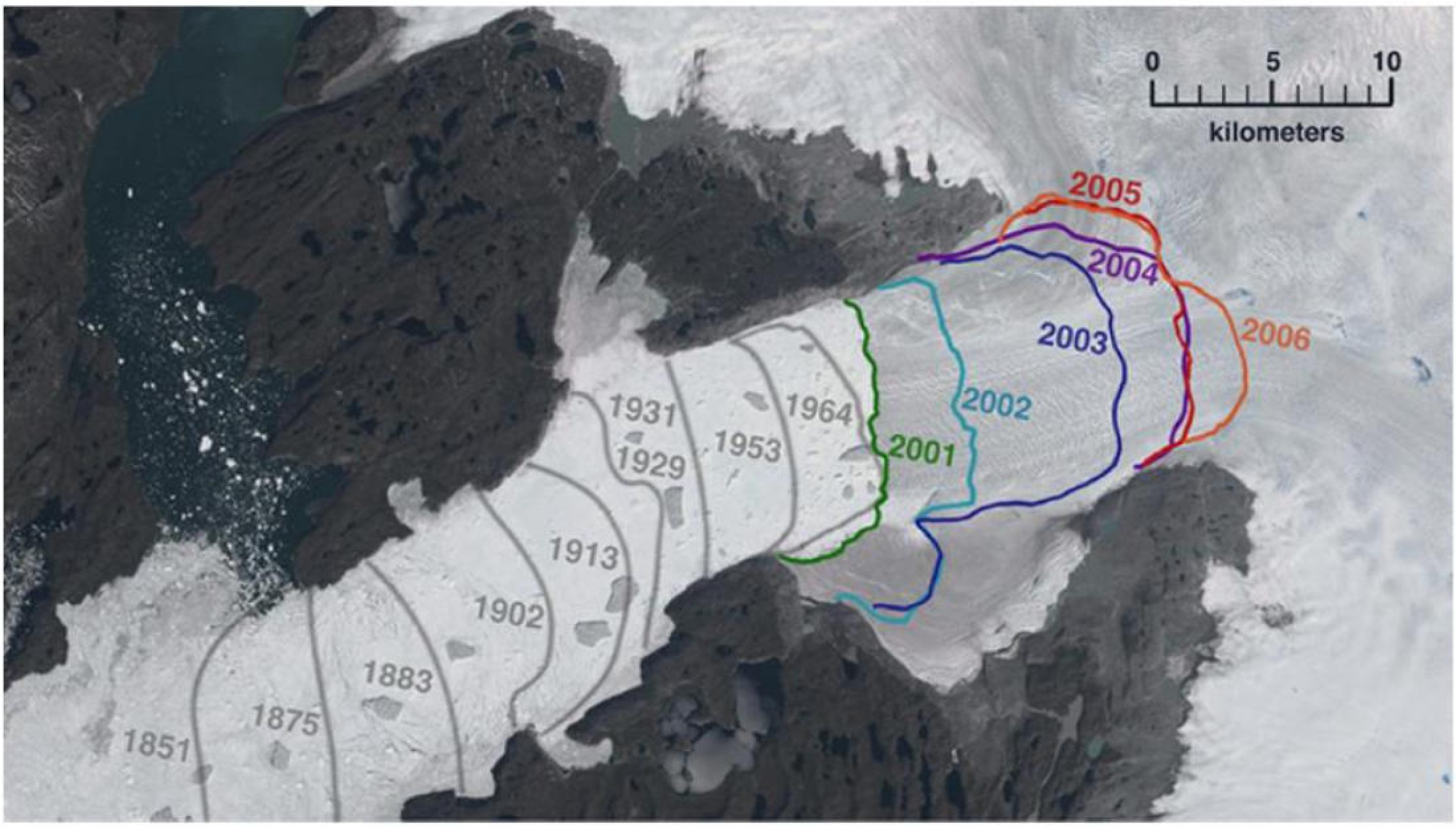
These increased temperatures have led to most glaciers shrinking at an alarming rate over the past few decades (Figure 5.9). Some have even vanished altogether. Melting glaciers and ice sheets are the biggest cause of rising sea levels. The Greenland and Antarctic ice sheets are the largest contributors of global sea level rise. On average, global sea levels are rising at 3.6 mm per year. By 2100, global sea levels are predicted to rise by 0.8 metres. Wildlife in high latitude areas is being affected. Polar bears in the Arctic are losing their sea ice habitat. This also has an impact on seals, the prey of polar bears, which rely on sea ice to raise their pups. Similar impacts can be seen in Antarctica where, in November 2022, early break up of sea ice in the Bellingshausen Sea killed 9000 emperor penguin chicks.

There are some potentially positive impacts of melting sea ice in polar regions. This could allow exploration of these areas for resources like oil, natural gas, and minerals. Ice-free summers in the Arctic Ocean, for example, could also open more efficient trading routes between North America and Eurasia.

When sea water gets warmer, **thermal expansion** occurs. This causes the volume of the water to increase. About half of the measured global sea level rise on Earth is from warming waters and thermal expansion. Low-lying countries such as Bangladesh, India, and the Netherlands, as well as low-lying island nations like Tuvalu, the Maldives, and Vanuatu are predicted to be most affected by rising sea levels. This will result in large areas becoming uninhabitable.

Increasing ocean temperatures means that marine plants and animals find it harder to survive. This may result in changes to species distribution. This could be both a positive and a negative, as fishing grounds change for different countries. For example, red mullet, sardines, and sea bass have all appeared with increasing frequency in fishermen's nets in the North Sea as fish migrate to cooler waters. However, fish like sardines, pilchards, and herring are shrinking in size due to warming oceans which will affect their ability to evolve and adapt. Coral bleaching due to increased sea temperatures can kill corals in tropical regions if warmer sea temperatures are sustained.

Figure 5.9: Muir Glacier, Alaska in 1941 (left) and 2004 (right)



Source A: An annotated photograph that shows the extent of Jakobshavn Isbræ, a glacier on the west coast of Greenland, between 1851 and 2006

TASK 1: Study Source A

a **Approximately how much did Jakobshavn Isbræ recede** between 1851 and 2006?

b **Which time period** shows the greatest rate of melting over the timescale shown?

c Explain the impacts of melting glaciers due to global warming. Consider both local and global consequences.

Weather patterns

Climate change affects weather patterns, causing more heatwaves, droughts, floods, and extreme weather events like tropical storms. **Heatwaves** are becoming increasingly common with climate change, and **cold snaps** are decreasing in frequency. Heatwaves put vulnerable people such as the elderly, young children, and people living in poverty at increased risk of heatstroke, which can result in deaths. During the summer of 2022, as many as 60 000 excess deaths were attributed to the climate change-related heatwave in Europe, with record-breaking temperatures of more than 47 °C. In the future, we will see more heatwaves and fewer periods of colder weather.

Climate change has also intensified the frequency and severity of **droughts** across the globe. Ethiopia, for example, has been badly affected by droughts since 2015 (impacting an estimated 24 million Ethiopians). Droughts can result in crop failure and **food insecurity**. The World Health Organisation estimates that 700 million people are at risk of being displaced as a result of drought by 2030. On the other hand, in the future, colder countries may experience higher crop yields because there will be a longer growing season due to increased temperatures. These changing patterns will affect the global food network (see Topic 10 for more on this).

Source B: Drought in Ethiopia has affected an estimated 24 million people since 2015

TASK 2: Study Source B

a Define the following key terms:

- **i** Glacial period
- **ii** Drought
- **iii** Climate change
- **iv** Global warming potential

b In pairs, using Source B, explain what the short-term and long-term impacts of droughts could be on people. Categorise the long-term impacts of drought as either social, economic, or environmental. You could capture this information in a table.

c In your pairs, discuss if you think men and women are affected differently by drought. Carry out research to supplement your discussion.

Climate change is also leading to more frequent and intense extreme weather events, such as tropical storms. For example, recent hurricanes in the Atlantic Ocean have had much higher wind speeds than previously. More intense tropical storms lead to increased loss of life, greater economic cost from damage, and interruptions to food and water supplies.

This warming trend also leads to changes in precipitation patterns, causing some regions to experience more intense and prolonged rainfall, increasing the risk of flooding. Climate change-induced flooding can have significant impacts on countries worldwide. In Bangladesh, for example, sea level rise and changing weather patterns have increased the risk of coastal and river flooding. This has affected millions and caused widespread damage. Increased urbanisation combined with intense rainfall events have amplified flood risk in other areas. For example, Houston, Texas, USA was flooded as a result of Hurricane Harvey in 2017, which had wind speeds up to 215 km per hour. Climate change also leads to more intense **monsoons**, for example in Kerala, India, leading to flash floods and landslides.

Figure 5.10: India is experiencing more intense monsoons which can result in flooding

Many areas around the globe will become more challenging places to live. People may have to leave their homes and be **displaced**. The Internal Displacement Monitoring Centre estimates that over 376 million people globally have been displaced by floods, windstorms, earthquakes, or droughts since 2008, with a record 32.6 million people in 2022 alone. Climate change will be an increasingly common reason why people migrate in the future. These people are called **climate refugees**.

TASK 3

'The impacts of climate change on people are worse than the impacts on the environment'.

Do you agree with this statement? Discuss in groups of four. Two people should argue for the impact on people being worse, and two people should argue for the impacts on the environment being worse. Remember to support your ideas with examples. After your discussion, come to a group conclusion and share your ideas with the rest of the class.

TASK 4

In pairs, conduct some research to create a fact file for a country that will benefit from the impacts of climate change, and a country that will suffer. Present your findings to the rest of the class and discuss if this is fair and what the solutions could be.